Gaussian Mixture Model

Our problem: heights without labels

Imagine measuring **1,000 heights**. You see two concentrations near 165 and 180 cm, but you do not know which group each person belongs to.

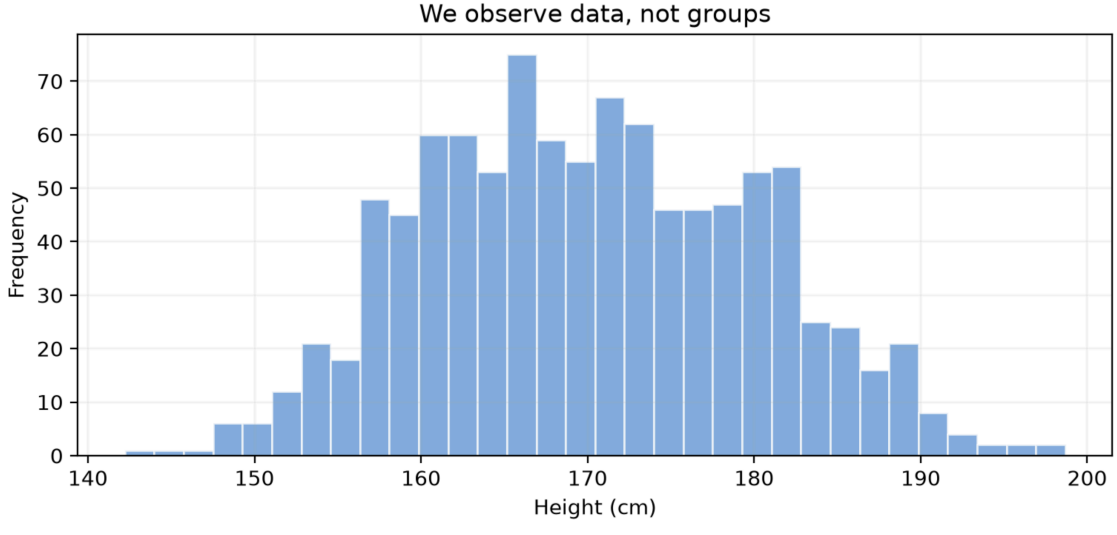
- **What do we observe?** Only height x.
- **What is missing?** The group label z.
- **What do we want?** Describe the groups and estimate membership.

Why start here? GMMs address precisely this problem of *hidden labels*.

One histogram hides two populations

The source group is hidden in each histogram. Change the distance between the means.

Distance between means (cm) 15



Why is one Gaussian sometimes insufficient?

A Gaussian describes **one center** and the spread around it.

If a population has several concentrations, one bell curve may represent it poorly.

Key idea: Combine several simple distributions instead of forcing one shape.

A single Gaussian may still describe overlapping or unimodal data; a mixture is not always needed.

The Gaussian

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

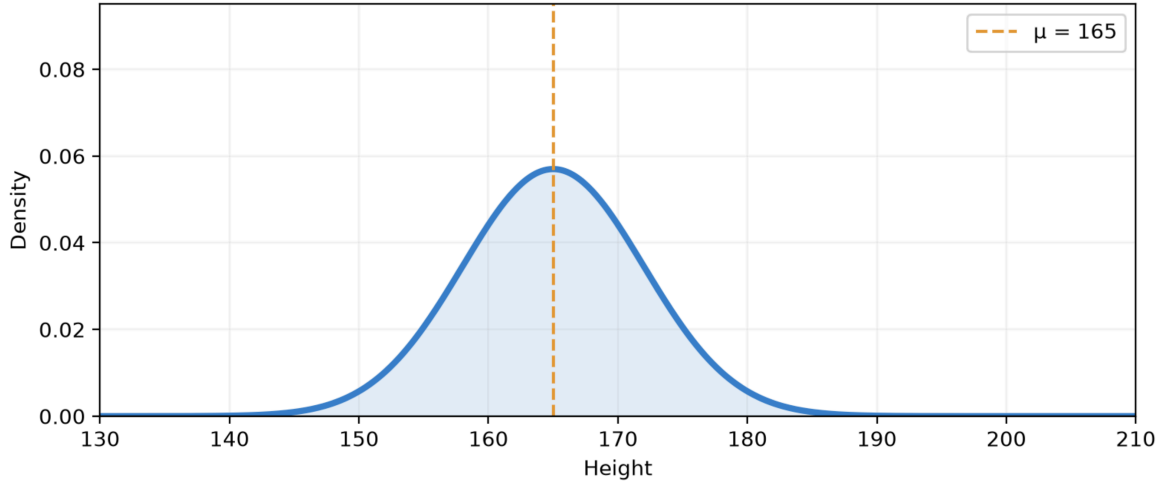
- μ (**mean**) moves the center.
- σ (**standard deviation**) controls the width.
- σ^2 is the **variance**.

Why the exponential? Density decreases as we move away from the mean. **Why the factor in front?** It normalizes the area under the curve to 1.

Mean μ

Move the mean: the curve shifts without changing width.

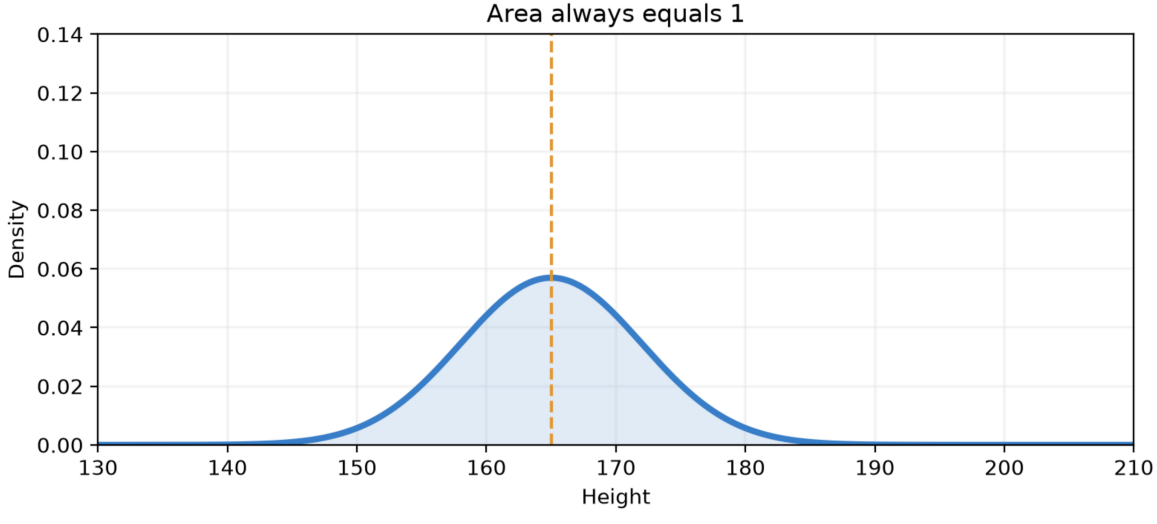
Mean μ (cm) 165



Spread σ

Change the standard deviation. Why is a wider bell curve also lower?

Standard deviation σ 7



Density is not the probability of an exact value

For a continuous variable, $P(X = 172) = 0$.

The curve gives a **density**. Probabilities come from the **area over an interval**:

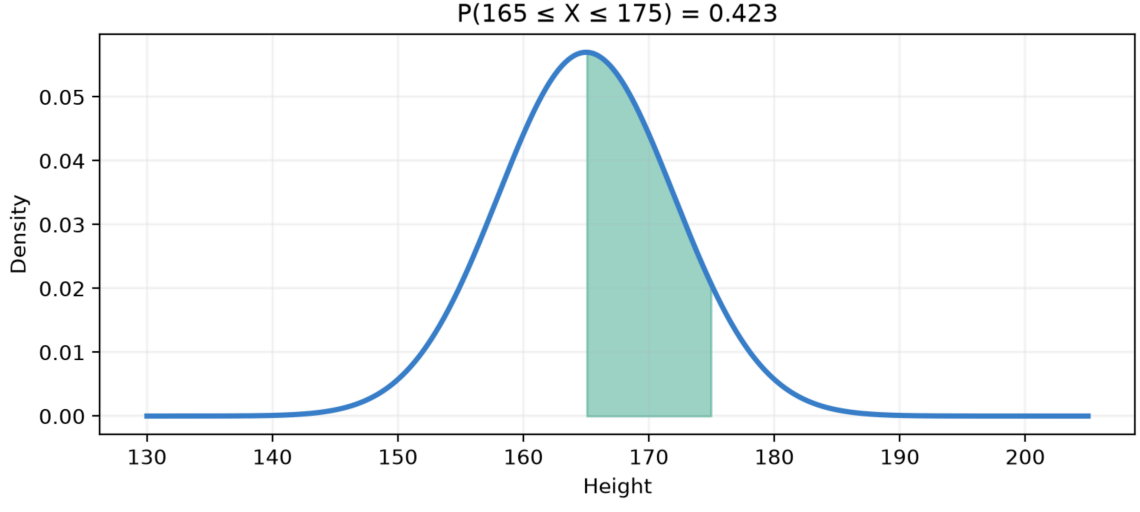
$$P(170 \leq X \leq 175) = \int_{170}^{175} f(x)\, dx.$$

Why do we need density? To compare how compatible a height is with different Gaussians. Never treat the curve's height as the probability of an exact value.

Probability = area under the curve

Choose the upper endpoint of an interval starting at 165 cm. The shaded area is the probability.

Upper endpoint (cm) 175



Introducing the GMM: a sum of weighted Gaussians

With two groups:

$$p(x) = \pi_1 \mathcal{N}(x \mid \mu_1, \sigma_1^2) + \pi_2 \mathcal{N}(x \mid \mu_2, \sigma_2^2)$$

Why a sum? A point may come from the first **or** the second group; we add the different ways it could be generated.

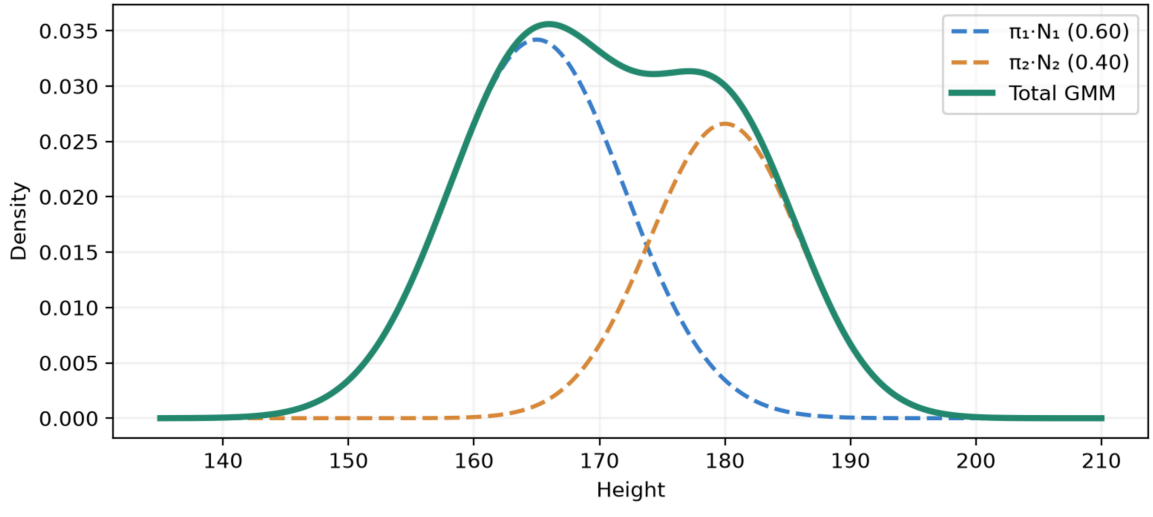
Why weights π_k? One group may occur more often than another.

$$\pi_k \geq 0, \qquad \sum_{k=1}^{K} \pi_k = 1.$$

Three curves: component 1, component 2, mixture

Change the blue group's weight. The green line is always the sum of the two weighted curves.

Group 1 weight (%) 60



General formula for K groups

$$\boxed{p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)}$$

- K: number of components.
- π_k: prior probability of choosing component k.
- μ_k: center of the component.
- Σ_k: variance (1D) or covariance (multiple dimensions).

Question: Must a GMM with K components have K peaks? **No.** Components can overlap and form a single peak.

How is an observation generated?

Step 1 — Draw the latent label z:

$$z \sim \text{Categorical}(\pi_1, \ldots, \pi_K).$$

Step 2 — Given the chosen label, draw the observation:

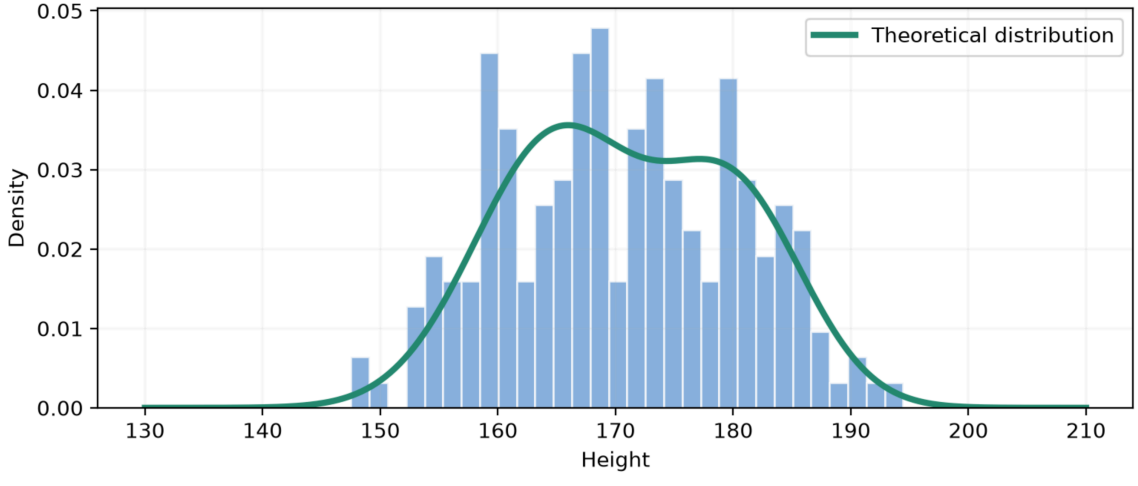
$$x \mid z = k \sim \mathcal{N}(\mu_k, \Sigma_k).$$

Why make this explicit? The model *generates the group first and then the data point*; in real data, we see only the point.

Generate data from the mixture

Increase the number of simulated observations. The latent groups start to emerge even though we see only their combined distribution.

N observations 200



Latent labels and soft clustering

In the dataset we know x_i, **not** z_i. Rather than immediately choosing a single label, we calculate:

$$P(z_i = k \mid x_i).$$

This is a *posterior probability*.

- **K-Means:** assigns each point to one group (hard clustering).
- **GMM:** assigns a probability to each group (soft clustering).

Why is this useful? Points in an overlap region remain uncertain instead of being forced into one group.

Bayes: from densities to responsibilities

$$\boxed{\gamma_{ik} = P(z_i = k \mid x_i) = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}}$$

Numerator: group frequency $\times$ compatibility with the observation.

Denominator: the contributions of **all** groups. It normalizes the result, so $\sum_k \gamma_{ik} = 1$.

Name: γ_{ik} is the *responsibility*: how strongly component k explains point i.

Bayes: a calculation for 172 cm

Use $\pi_1 = 0.6, \mu_1 = 165, \sigma_1 = 7; \pi_2 = 0.4, \mu_2 = 180, \sigma_2 = 6$.

$$f_1(172) \approx 0.03458, \quad f_2(172) \approx 0.02734$$

$$a_1 = 0.6 \cdot 0.03458 \approx 0.02075, \quad a_2 = 0.4 \cdot 0.02734 \approx 0.01094$$

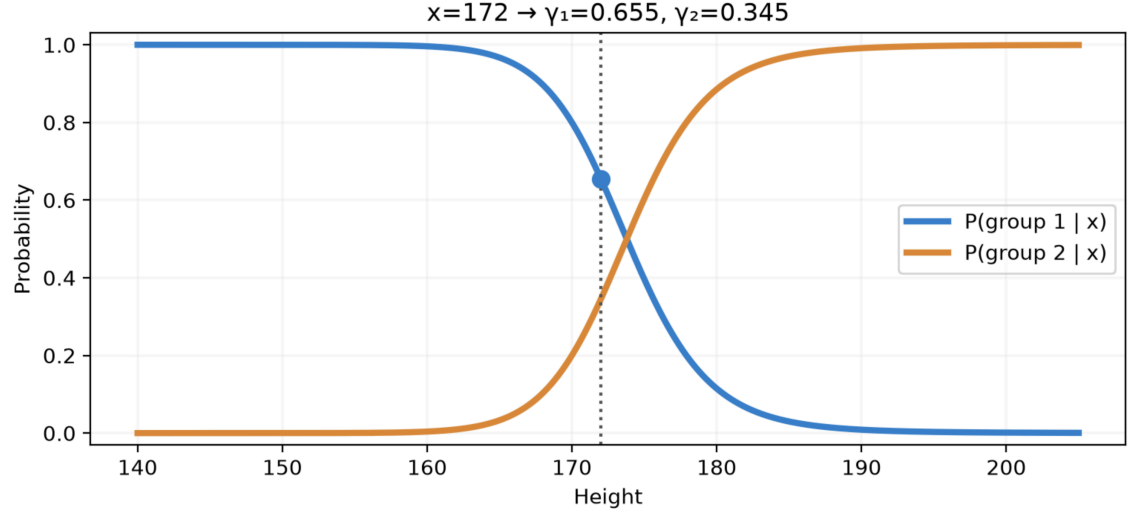
$$\gamma_1 = \frac{a_1}{a_1 + a_2} \approx 0.655, \quad \gamma_2 \approx 0.345.$$

Why not just compare distances from the means? Spread and weights matter too.

Responsibility

Move the height slider. Membership may be uncertain in overlap regions; responsibilities change toward the tails.

Height x 172



The learning problem

So far we have known the parameters. With real data, we must estimate them:

$$\theta = \{\pi_k, \mu_k, \Sigma_k\}_{k=1}^{K}.$$

The observed data are $X = \{x_1, \ldots, x_N\}$.

Goal: Choose parameters that make the observed data as plausible as possible (**maximum likelihood**).

But how can we estimate the components without knowing z_i?

Likelihood: why a product and a sum?

Assuming independent observations:

$$L(\theta) = \prod_{i=1}^{N} \underbrace{\left[\sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k) \right]}_{\text{density of point } i}.$$

- **Product across data points:** Each observation contributes to the overall plausibility.
- **Sum across components:** Each point could come from any of the K Gaussians.

Do not confuse *likelihood* with a probability distribution over parameters: here the parameters are values to optimize.

Why use log-likelihood?

$$\boxed{\ell(\theta) = \sum_{i=1}^{N} \log \left(\sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k) \right)}$$

Why take the logarithm? It turns a product into a sum and avoids numerical underflow.

Remaining difficulty: A **sum over components** remains inside the logarithm, so the optimum cannot be found with the usual sample-mean formula.

Solution: Alternate between estimating membership and updating parameters.

EM: the cycle that solves the problem

E-step (Expectation) — With fixed parameters, calculate $\gamma_{ik} = P(z_i = k \mid x_i)$.

M-step (Maximization) — With fixed responsibilities, update π_k, μ_k, and Σ_k.

Repeat until convergence.

Why does this work? If we knew membership, estimating the Gaussians would be easy. If we knew the Gaussians, estimating membership would be easy. EM uses this alternation.

Initializing EM

Example: $\mu_1 = 160$, $\mu_2 = 185$, $\sigma_1 = \sigma_2 = 10$, $\pi_1 = \pi_2 = .5$.

These are **initial guesses**, not final values.

Why does initialization matter? Likelihood is not necessarily convex; EM can settle on different local solutions.

In practice: Initialize with K-Means and try multiple runs (`n_init`).

E-step: recalculate every responsibility

For each point i and Gaussian k:

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}.$$

What changes? The responsibilities. **What stays fixed?** The model parameters.

Example row: $x_i = 172$, $\gamma_{i1} = .655$, $\gamma_{i2} = .345$.

Responsibilities in each row always sum to 1.

M-step: updating the mean

Define the effective number of points in the group:

$$N_k = \sum_{i=1}^{N} \gamma_{ik}.$$

The new mean is:

$$\boxed{\mu_k^{new} = \frac{\sum_i \gamma_{ik} x_i}{N_k}}$$

Why a weighted mean? A point that strongly belongs to a component influences its center more; an uncertain point influences it less.

Example: $x = (160, 170, 180)$, $\gamma_{\cdot 1} = (.9, .6, .1)$ gives $\mu_1 = 165$.

M-step: weighted variance

In 1D:

$$\boxed{(\sigma_k^2)^{new} = \frac{\sum_i \gamma_{ik}(x_i - \mu_k^{new})^2}{N_k}}$$

Why square the deviations? To measure spread without positive and negative deviations canceling out.

Why use weights? Variance should mainly describe the points assigned to that component.

With $x = (160, 170, 180)$, $\gamma = (.9, .6, .1)$, and mean 165: variance is 37.5 and $\sigma \approx 6.12$.

M-step: updating the weights

$$\boxed{\pi_k^{new} = \frac{N_k}{N} = \frac{1}{N}\sum_i \gamma_{ik}}$$

Why this formula? N_k counts the points *effectively* assigned to the group. Dividing by N gives its overall share.

Example: $N = 3$, $N_1 = .9 + .6 + .1 = 1.6$.

$$\pi_1^{new} = 1.6/3 \approx 0.533.$$

Check: $\sum_k \pi_k^{new} = 1$ automatically.

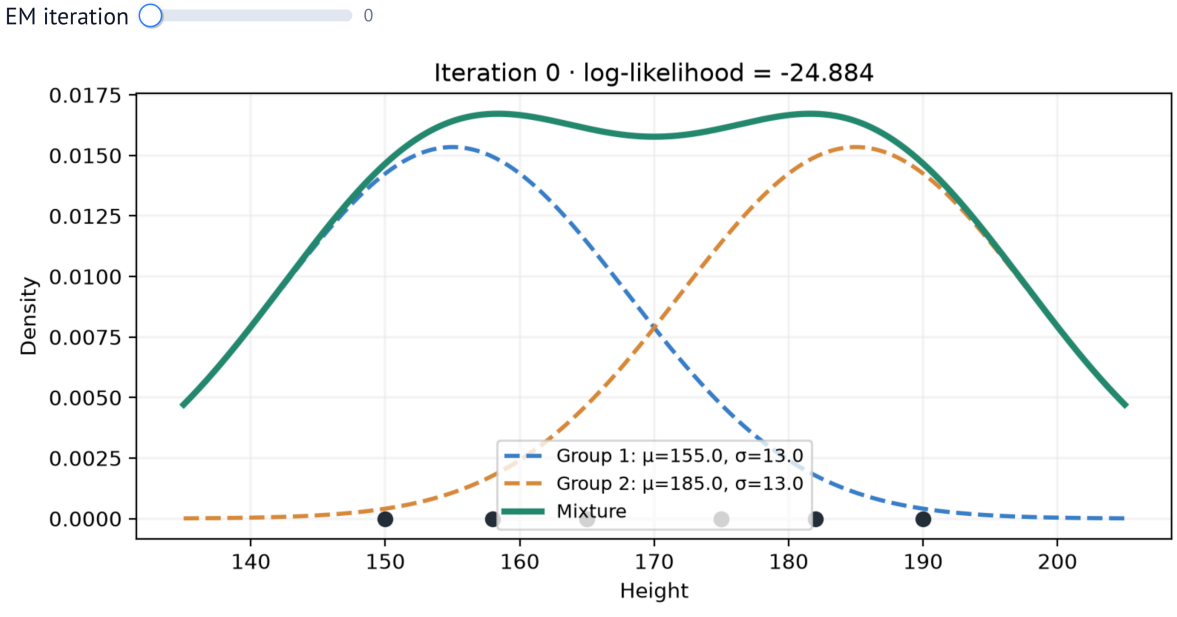
Complete example: one EM iteration

Data: $x = (160, 170, 180)$; assumed responsibilities for group 1: $(.9, .6, .1)$.

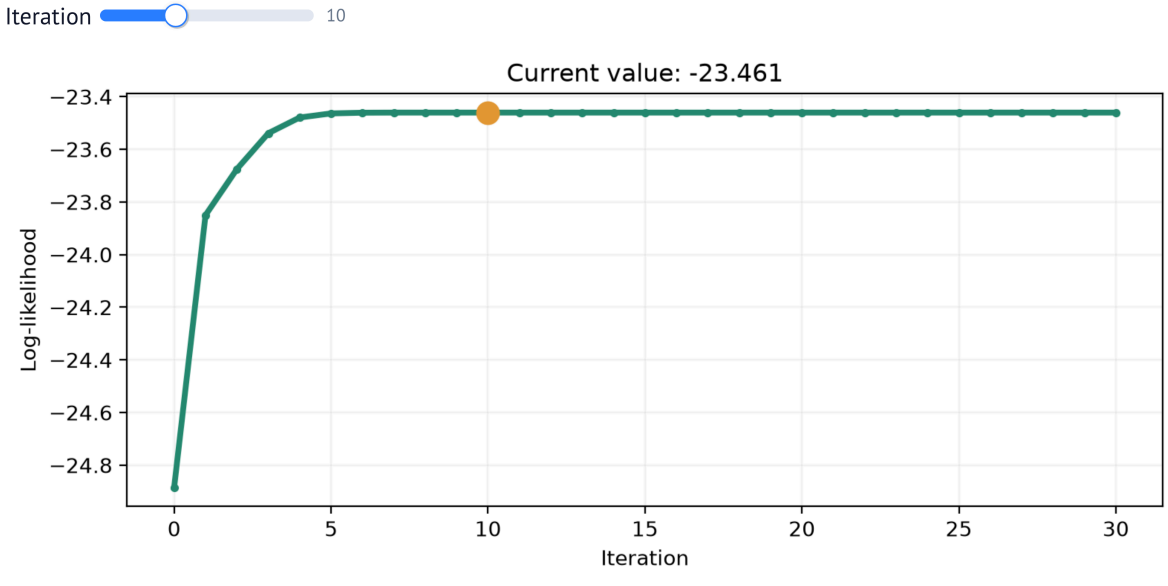
1. **E-step already done:** We have $\gamma_{\cdot 1}$, and $\gamma_{\cdot 2} = (.1, .4, .9)$.
2. **Effective counts:** $N_1 = 1.6$, $N_2 = 1.4$.
3. **Means:** $\mu_1 = 165$, $\mu_2 = (.1 \cdot 160 + .4 \cdot 170 + .9 \cdot 180)/1.4 \approx 175.71$.
4. **Variances:** Each is a weighted mean of squared deviations.
5. **Weights:** $\pi_1 \approx .533$, $\pi_2 \approx .467$.

On the next iteration, recalculate responsibilities with the **new** parameters.

Each slider position shows an E-step followed by an M-step.



EM increases log-likelihood



Convergence and limitations of EM

EM stops when the improvement in log-likelihood becomes very small or it reaches the iteration limit.

Property: Standard EM iterations do not decrease log-likelihood.

But keep in mind:

- Convergence does **not** guarantee a global maximum.
- A component may collapse onto one point ($\sigma^2 \to 0$), causing degeneracy.
- Covariance regularization and multiple initializations help.

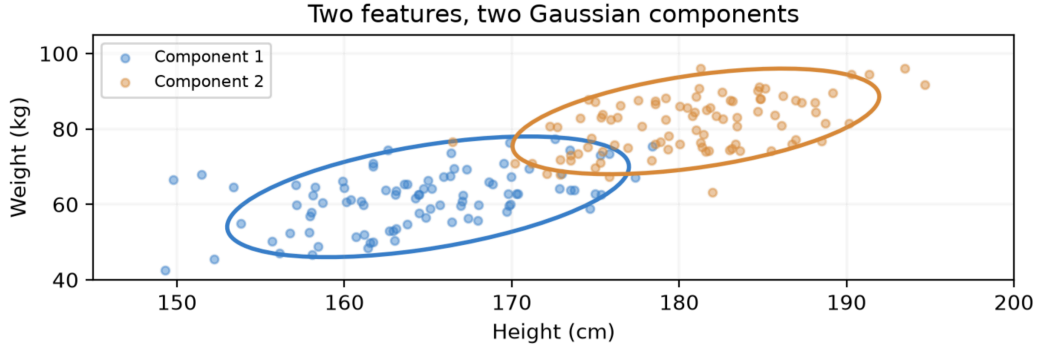
From a line to a plane: multidimensional data

With height **and** weight, each observation becomes a vector:

$$x_i = \begin{bmatrix} \text{height}_i \\ \text{weight}_i \end{bmatrix}, \quad \mu_k = \begin{bmatrix} \mu_{h,k} \\ \mu_{w,k} \end{bmatrix}.$$

Components are no longer simple curves: they become Gaussian surfaces with **elliptical contours**.

Why make this extension? Real data often require clustering with several features at once.



What does a covariance matrix add?

$$\Sigma_k = \begin{bmatrix} \sigma^2_{h,k} & \mathrm{Cov}(h,w) \\ \mathrm{Cov}(h,w) & \sigma^2_{w,k} \end{bmatrix}$$

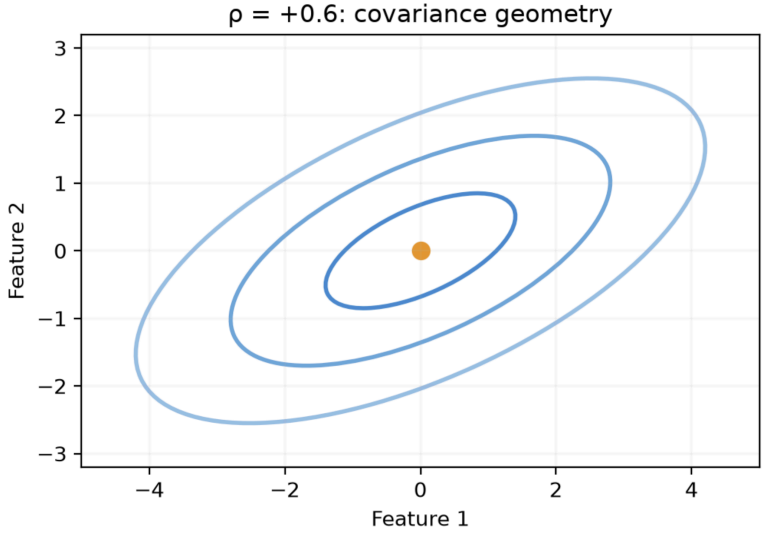
- Diagonal: spread of each feature.
- Off-diagonal: how two variables increase or decrease together.

Why does it matter? A cluster may be elongated and rotated; correlations determine its orientation.

Ellipses and covariance

Change the correlation. The ellipses rotate while keeping the same center.

Correlation ρ 0.6



The multivariate Gaussian

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right]$$

- d: dimension of the vector.
- $|\Sigma|$: volume/normalization factor.
- Σ^{-1}: accounts for scale and correlation.
- The quadratic term is the **squared Mahalanobis distance**.

Why not Euclidean distance? Moving along a direction with large spread is less surprising than moving along a narrow direction.

Multidimensional M-step

The mean is still weighted:

$$\mu_k^{new} = \frac{\sum_i \gamma_{ik} x_i}{N_k}.$$

The variance becomes a matrix:

$$\boxed{\Sigma_k^{new} = \frac{\sum_i \gamma_{ik}(x_i - \mu_k^{new})(x_i - \mu_k^{new})^T}{N_k}}$$

Why the outer product? It captures both the variance along each dimension and covariance between dimensions.

How many Gaussians should we choose?

We do not always know K. Too few components may underfit; too many add unnecessary complexity.

BIC balances fit and number of parameters:

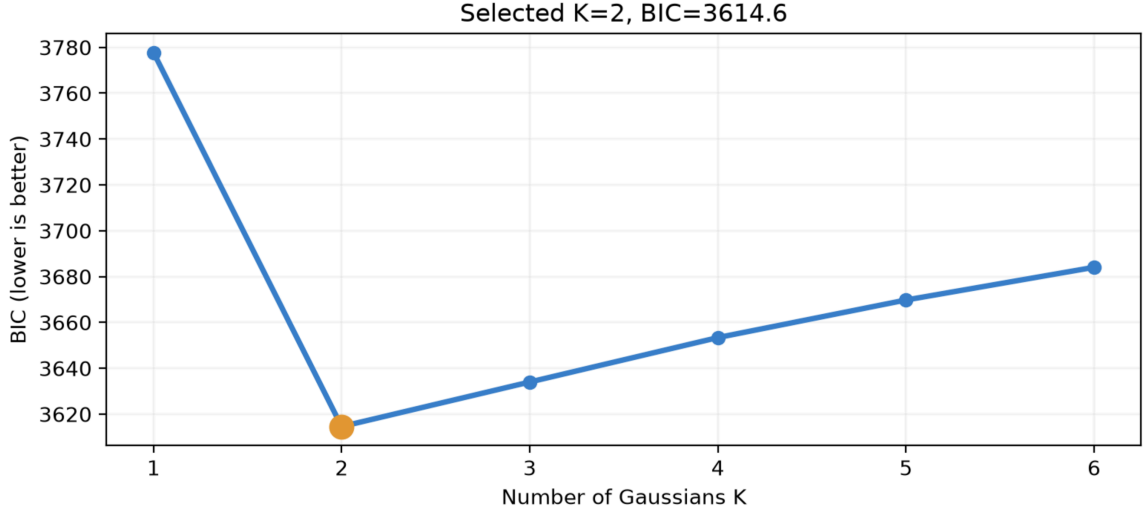
$$\mathrm{BIC} = -2\ell(\hat{\theta}) + p \log N.$$

Why is lower better? The first term rewards high likelihood; the second penalizes complexity. We test several values of K.

BIC: compare $K = 1, \ldots, 6$

For synthetic data, the BIC minimum suggests balanced model complexity, not necessarily the number of real-world categories.

Selected K 2



When should you use a GMM?

Useful when:

- You want membership probabilities (soft clustering).
- Groups overlap.
- Groups are elliptical and have different variances.
- You want to estimate a density or identify poorly fitting data points.

Limitations: Choosing K, sensitivity to initialization, Gaussian component assumptions, and possible covariance degeneracy.

Remember: A mathematical component is not automatically a real-world category.

Three formulas to remember

Model:

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k).$$

E-step:

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}.$$

M-step (mean):

$$\mu_k^{new} = \frac{\sum_i \gamma_{ik} x_i}{\sum_i \gamma_{ik}}.$$

Logical sequence: Parameters → membership → new parameters → repeat.

Solutions and conclusion

1. The weights are probabilities of choosing the groups.
2. Gaussians can overlap; the labels are hidden.
3. The E-step updates **only** responsibilities.
4. Data points should contribute in proportion to their compatibility with the cluster.
5. **No:** EM can converge to local maxima or degenerate solutions.
6. To describe spread and correlation across dimensions.

Takeaway: A GMM is a probabilistic generative model. EM learns it by alternating **inference** about hidden labels and **estimation** of parameters.